SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: MLA03
COURSE NAME: Reinforcement learning for Environment and Agent

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COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10 **Total Marks: 50**

Annexure A

Simulation-Based Assessment Question

Duration: 1hr

1. Simulate a **Hierarchical Reinforcement Learning (HRL)** agent for autonomous warehouse navigation.

   (5 Marks)

2. Implement a **Hierarchical Abstract Machine (HAM)** to control a robot performing sequential tasks.

   (5 Marks)

3. Develop the **MAXQ Value Decomposition** algorithm for a taxi navigation environment.

   (5 Marks)

4. Simulate a **Meta-Reinforcement Learning** agent that rapidly adapts to multiple GridWorld environments.

   (5 Marks)

5. Design a **Multi-Agent Reinforcement Learning (MARL)** system for cooperative robot path planning.

   (5 Marks)

6. Implement a **competitive Multi-Agent RL** environment for autonomous drone navigation.

   (5 Marks)

7. Simulate a **Partially Observable Markov Decision Process (POMDP)** for autonomous vehicle navigation under limited visibility.

(5 Marks)

8. Design an **ethical Reinforcement Learning** simulation for autonomous decision-making while satisfying safety constraints.

(5 Marks)

9. Develop an RL-based **adaptive traffic signal control** system and evaluate its performance.

10. Simulate an RL-based **smart energy management system** for optimizing power distribution in a microgrid.. **.**

(5 Marks)

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively